



Immunomodulatory effects of amisulpride on mammalian macrophages

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Abstract

Background The relationship between psychiatric disorders and the immune system has become increasingly recognized, with inflammatory pathways implicated in disease pathogenesis, therapeutic response, and progression. Amisulpride, an atypical antipsychotic preferentially acting as an antagonist on postsynaptic D2/D3 receptors, has been widely studied for its psychiatric effects, whereas its immunomodulatory properties remain largely unexplored.

Aim We investigated the potential immunomodulatory effects of amisulpride on J774.2 macrophage cells.

Methods J774.2 macrophage cells were treated with amisulpride at concentrations of 1, 5, and 10 µg/mL for 24 hours. Cytotoxicity was assessed, and proinflammatory cytokine production (TNF-α, GM-CSF, IL-6, IL-12p40) was measured by ELISA under both basal and LPS-induced inflammatory conditions.

Results Amisulpride demonstrated no cytotoxicity at any concentration. Under non-inflammatory conditions, cytokine levels remained unchanged. In contrast, upon LPS-induced inflammation, amisulpride elicited a dose-dependent anti-inflammatory response, significantly reducing cytokine secretion. **Conclusion:** Amisulpride exhibits immunomodulatory activity under inflammatory conditions. These findings suggest a potential anti-inflammatory mechanism that warrants further investigation.

Keywords Amisulpride · Anti-inflammatory · Antipsychotic · Immunomodulation

Introduction

Antipsychotic drugs are agents used to manage the symptoms of psychotic disorders, bipolar disorder, and various other psychiatric diseases. Amisulpride is a benzamide-derived antipsychotic frequently used in the treatment of schizophrenia, a psychotic disorder with an estimated lifetime prevalence of approximately 0.4% [1, 2]. It exerts its therapeutic effects primarily by reducing dopaminergic transmission through D2/D3 receptor antagonism at high doses, particularly in the limbic system. At low doses, it enhances dopamine transmission by blocking presynaptic D2/D3 receptors [3].

Proinflammatory cytokines are known to increase the production of reactive oxygen and nitrogen species through the activation of macrophages and neutrophils, whereas certain cytokines can affect the central nervous system by modulating the activity of monoaminergic systems [4–6]. When examining the effects of dopamine, a monoaminergic neurotransmitter, on macrophages, studies have shown

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that it increases the production of inflammatory cytokines (IL-6, IL-1 β , TNF- α) and chemokines (CCL2, CXCL8, CXCL9, CXCL10), while decreasing the production of the anti-inflammatory cytokine IL-10 [7, 8]. Elevated dopamine levels stimulate low-affinity dopamine receptors, leading to anti-inflammatory effects in immune cells. In contrast, reduced dopamine levels activate high-affinity receptors, thereby triggering an inflammatory response [9, 10]. In this context, the immunomodulatory effects of antipsychotic drugs that influence the relationship between dopamine and the inflammatory system have attracted considerable interest.

Several investigations have demonstrated a relationship between increased white blood cell count, monocytosis, and the worsening of psychotic symptoms, particularly in patients with first-episode psychosis (FEP) [11, 12]. Studies on cytokines have also supported the relationship between the immune system and psychotic disorders, opening a broad area of research. The significance of examining the immunomodulatory effects of psychotropic drugs on proinflammatory cytokines is further highlighted by recent findings [13, 14]. Evidence indicates that high serum levels of IL-6 during childhood are associated with future risks of depression and psychosis [15]; the excessive expression of IL-6 in the central nervous system contributes to chronic inflammation [16]; the negative correlation between IL-6 and BDNF in FEP patients [17]; and the role of elevated IL-6 levels in cognitive decline in schizophrenia [18] collectively support the connection between cytokines and psychotic disorders. Another study suggested that the inhibitory effects of antipsychotic drugs on the IL-6/STAT3 signaling pathway in schizophrenia patients may contribute to the improvement of psychotic symptoms [19]. A meta-analysis reported that IL-2 and IL-6 levels decreased following antipsychotic treatment in FEP patients, suggesting their potential role as biomarkers reflecting treatment response [20]. Another meta-analysis comparing cytokines has shown that IL-6 and TNF- α levels are higher in both FEP patients and chronic schizophrenia patients compared to controls: in particular, TNF- α , IL-6, and IL-12 levels were found to be higher in the chronic patient group experiencing exacerbations compared to controls [21]. Conflicting findings exist in the literature, with some studies indicating both increases and decreases in IL-12 levels following treatment [22, 23]. Moreover, there are studies drawing attention to the relationship between the IL-12p40 monomer, a shared subunit of IL-12 and IL-23, and psychosis [24, 25].

The fact that antipsychotic drugs can not only reduce psychotic symptoms but also alter cytokine levels indicates that their immunomodulatory effects might contribute to treatment. Considering this possibility, it is reasonable to infer that further studies are required to resolve the uncertainties

surrounding these processes. However, the immunomodulatory effects of the antipsychotic agent amisulpride remain poorly understood. This study aimed to examine the immunomodulator impact of amisulpride on mammalian macrophages at various concentrations, both in the presence and absence of LPS-induced inflammation.

Methods

Cell culture maintenance

The J774.2 macrophage cell line was cultured in complete Roswell Park Memorial Institute 1640 (RPMI 1640) medium supplemented with 10% FBS, 1% penicillin–streptomycin, and sodium pyruvate at 37 °C under 5% CO₂. The culture medium was changed every four days, and the cells were passaged upon reaching 80% confluency [26].

Cytotoxicity assessment of amisulpride

The effect of amisulpride on cell viability was assessed using trypan blue staining. A 1 mg/mL suspension of amisulpride (CAS: 71675–85-9) was prepared in distilled water by high-speed vortexing and filtered through a 0.22 μ m sterile membrane. The filtrate was diluted in culture medium to nominal concentrations of 1, 5, and 10 μ g/mL. Cells were seeded in a 24-well plate at a density of 3,000 cells per well. After 24 h of incubation to allow cell attachment, the culture medium was replaced with different control and treatment conditions. Cells were then incubated for another 24 h under one of the following conditions: culture medium only (control), LPS alone, amisulpride at 1, 5, or 10 μ g/mL, or a combination of LPS and Amisulpride at the same concentrations.

Amisulpride treatment and macrophage activation by lipopolysaccharide (LPS) stimulation

The immunoregulatory effect of Amisulpride treatment on J774.2 macrophages was tested in vitro, under either LPS-stimulated or LPS-free conditions. Briefly, J774.2 cells were plated in 24-well plates at a density of 1×10^6 cells per plate in complete RPMI medium and incubated under normal culture conditions for 24 h before treatment to ensure attachment. For LPS-mediated stimulation of cytokine release, 1 μ L/well LPS (1 mg/mL, *Salmonella Minnesota* R595, Enzo Life Sciences) was added to the cell culture medium. Amisulpride (CAS: 71,675–85-9), prepared as previously described, was applied to J774.2 macrophages at nominal concentrations of 1, 5, and 10 μ g/mL, in the presence or absence of LPS-containing culture medium. After 24 h of amisulpride treatment, with or without LPS stimulation,

the proinflammatory cytokine release response was measured via ELISA. Additionally, cells exposed only to LPS-containing medium, as well as those cultured in complete medium, were analyzed to assess both basal and stimulated cytokine secretion levels. After treatment, culture supernatants were collected from 24-well plates and centrifuged at 2000 rpm to remove cellular debris. The clarified supernatants were distributed into the wells of a 96-well plate.

Quantification of cytokine release by enzyme-linked immunosorbent assay (ELISA)

An Enzyme-Linked Immunosorbent Assay (ELISA) was used to quantitatively assess the levels of proinflammatory cytokines TNF- α (cat number: 555212), GM-CSF (cat number: 555126), IL-6 (cat number: 555183), and IL-12p40 (cat number: 555220) in macrophages, as previously described [27].

Initially, a coating solution was prepared by diluting hamster anti-mouse antibody in bicarbonate buffer (0.5 μ g/mL, pH 9.5) at a 1:1000 ratio. Then, 100 μ L of this solution was added to each well of a 96-well plate. The plates were incubated at 4°C overnight to ensure optimal antibody binding. The next day, wells were emptied and washed three times with 200 μ L of 0.05% Tween 20 in PBS solution (PBS-T, Cat. no: 11332465001) to remove unbound capture antibody. Then, 200 μ L of 1% BSA in PBS was added to block non-specific antibody binding for 3 h at room temperature to enhance blocking efficiency. Following the incubation, the blocking buffer was removed, and wells were thoroughly rinsed three times with PBS-T.

The samples were diluted 1:1000 in sample buffer, and standards were prepared according to the manufacturer's instructions. Then, 100 μ L of each sample was transferred into the coated wells. The standard curve was generated by serial dilutions of recombinant cytokine standards. The plate was incubated at room temperature (RT) for three hours to allow cytokine binding. Then, the plates were rinsed three times with PBS-T.

For the detection step, biotinylated anti-mouse cytokine antibodies targeting IL-6, GM-CSF, TNF- α , and IL12p40 were diluted to a final concentration of 0.5 μ g/mL in PBS containing 10% FBS. Then, 100 μ L of the antibody solution was added to each well and incubated at RT for one hour. The plates were aspirated and washed three times with PBS-T to remove excess detection antibodies. Streptavidin-HRP solution (100 μ L) was added to each well and incubated for 2 h at RT.

Next, 100 μ L of TMB substrate solution (BD OptEIA) was added to each well and incubated in the dark for approximately 30 min until the desired color intensity developed. To terminate the reaction, 50 μ L of 1 M sulfuric acid was added

to each well. The absorbance of the samples was measured at 450 nm using a microplate reader. Cytokine concentrations were quantified by normalizing the absorbance values against the standard curve, as previously described [27].

Statistical analysis

Statistical analyses were conducted using GraphPad Prism version 10.3.1 (San Diego, California, USA). To test the effect of Amisulpride on viability, the normality of the data was confirmed using the Shapiro–Wilk test. One-Way ANOVA was then performed to assess variability among the groups, followed by Dunnett's multiple comparisons test to compare each treatment group with the control. To quantitatively assess the effect of Amisulpride on cytokine levels, the normality of the data was evaluated using the Shapiro–Wilk test. After confirming that the data were normally distributed, One-Way ANOVA was conducted to compare cytokine levels among treatment groups, followed by Tukey's multiple comparisons test to identify significant pairwise differences. *p*-values below 0.001 were considered statistically significant. Each condition was tested in triplicate.

Results

This study examines the modulation of the pro-inflammatory cytokine profile in J774.2 macrophage cells upon Amisulpride treatment. Cells were cultured with three different concentrations of Amisulpride (1, 5, and 10 μ M) for 24 h. The conditions were also repeated under LPS-induced inflammatory conditions to capture the potential anti-inflammatory effect of Amisulpride.

Cell viability was assessed using trypan blue staining under the same experimental setup that would later be applied for ELISA-based cytokine assessment to ensure that any observed changes in the cytokine profile were not due to cytotoxic doses of Amisulpride (Fig. 1). The findings showed that Amisulpride, at the tested doses, did not have a significant impact on cell viability within 24 h, either in the presence or absence of LPS-induced inflammatory conditions.

Amisulpride treatment for 24 h was evaluated for its potential influence on cytokine secretion in J774.2 macrophages (Fig. 2). However, no detectable change was recorded in the ELISA results. TNF- α , GM-CSF, IL-6, and IL12p40 cytokine levels were measured, but no detectable elevation was observed compared to control groups under normal culture conditions. In contrast, the addition of 1 μ L (1 mg/mL) LPS significantly stimulated the secretion of pro-inflammatory cytokines TNF- α , GM-CSF, IL-6, and

Fig. 1 Cell viability assay of J774.2 macrophages treated with different concentrations of amisulpride. *p*-values smaller than 0.001 were considered statistically significant. Data are presented as mean $\pm$ SD

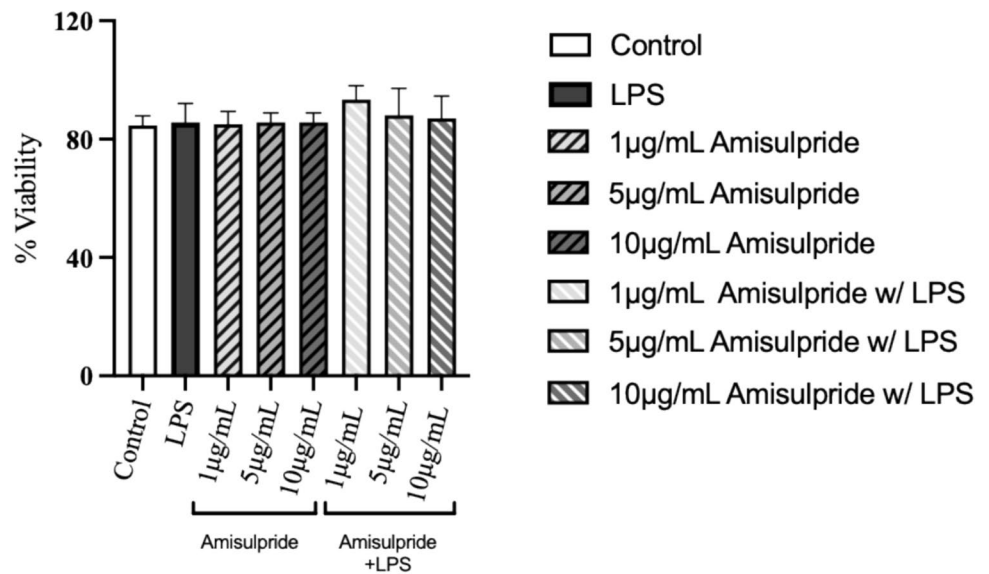


Fig. 2 Quantitative analysis of pro-inflammatory cytokine secretion -**A** TNF- α , **B** IL-6, **C** GM-CSF, and **D** IL-12p40- was conducted in J774.4 macrophages following amisulpride treatment (1, 5, and 10 μ M) or LPS (1 μ L, 1 mg/mL) alone. All conditions were performed in triplicate ($p < 0.05$). Data represent mean $\pm$ SD. ** $p < 0.001$ versus control

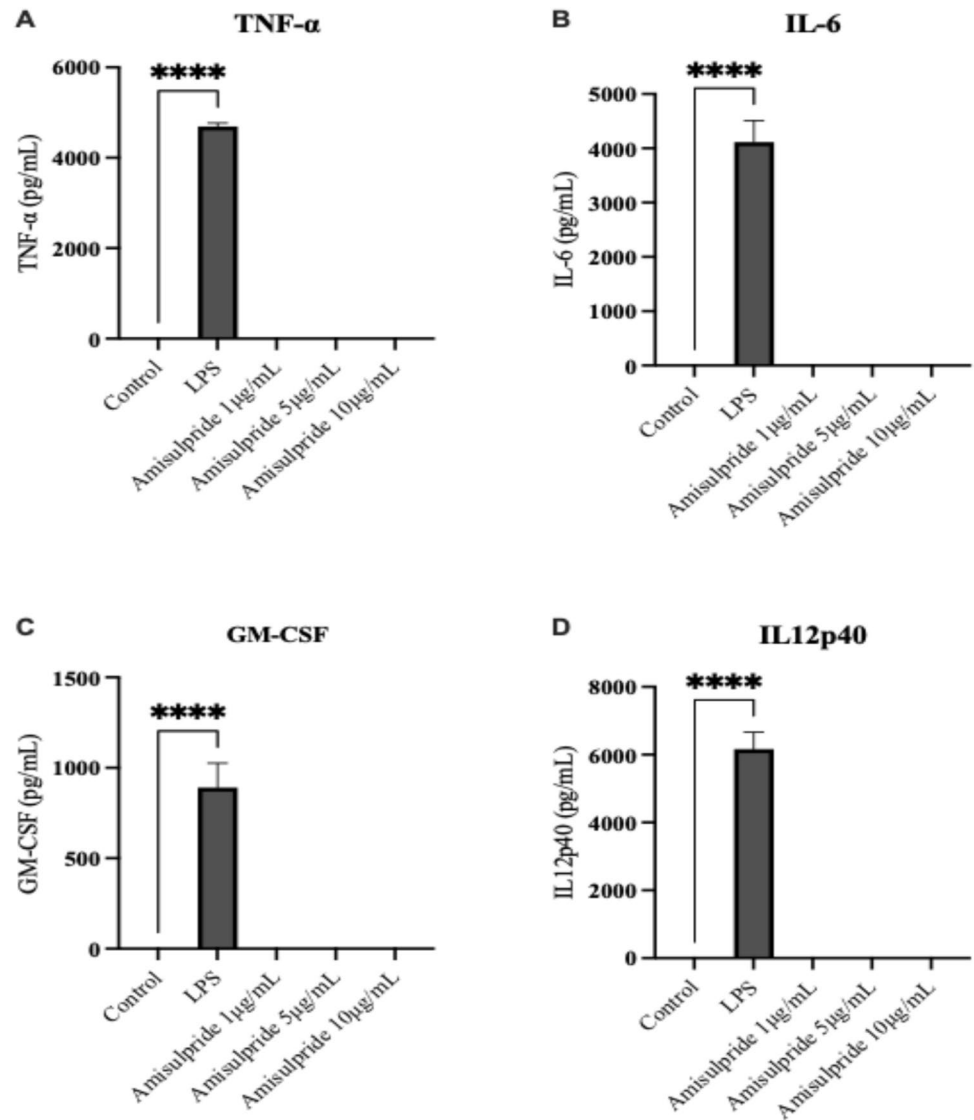
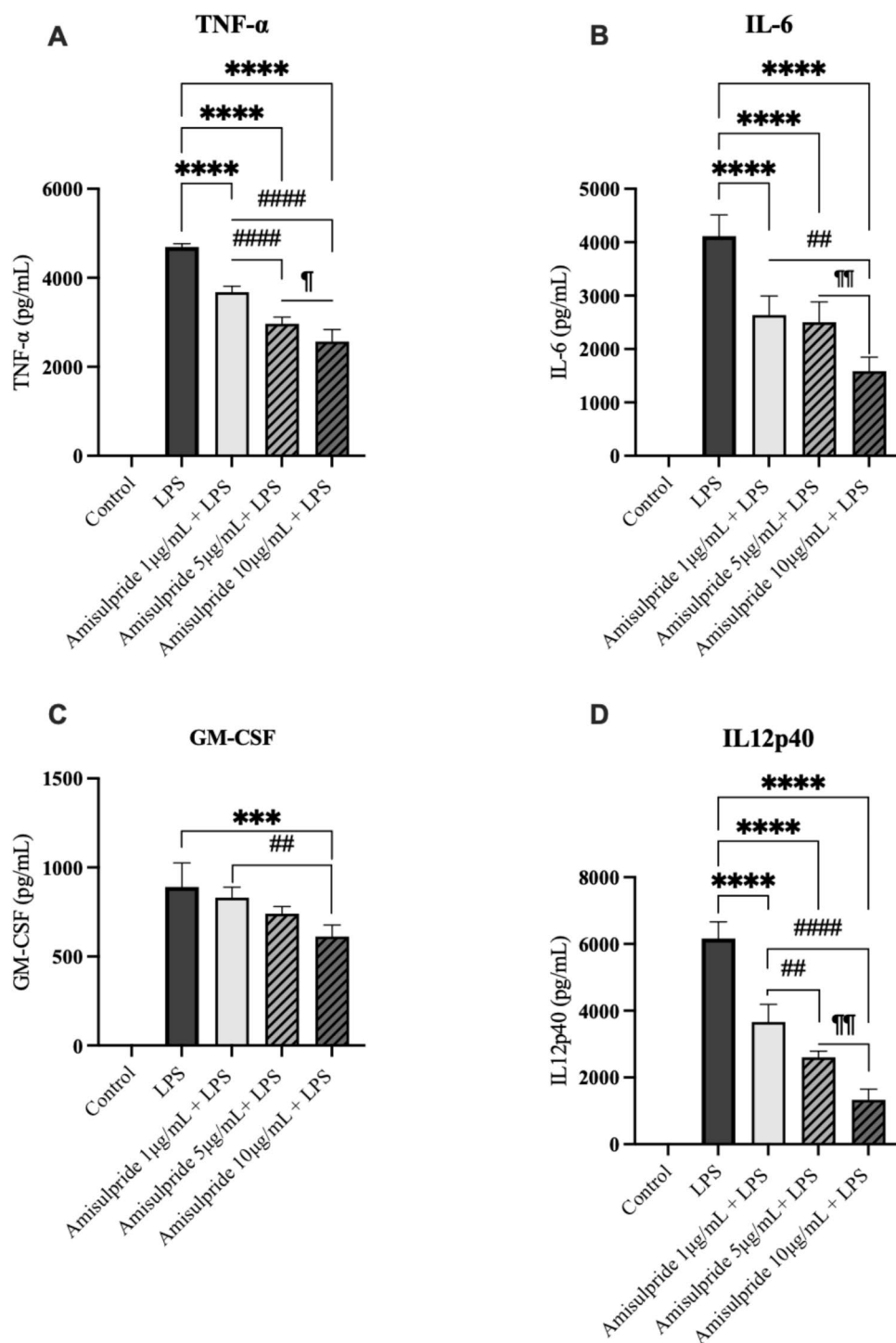


Fig. 3 Quantitative analysis of pro-inflammatory cytokine secretion following amisulpride treatment (1, 5, and 10 μ M) in J774.4 macrophage cells. **A** TNF- α , **B** IL-6, **C** GM-CSF, and **D** IL-12p40 levels were measured under LPS-induced conditions. All conditions were performed in triplicate ($p < 0.05$). Data represent mean $\pm$ SD. ** $p < 0.001$, * $p < 0.0001$ versus LPS; ## $p < 0.01$, #### $p < 0.0001$ versus 1 μ g/mL Amisulpride+LPS; $p < 0.05$, $p < 0.01$ versus 5 μ g/mL Amisulpride+LPS



IL12p40 in J774.2 cells. These observations suggest that Amisulpride, at the tested concentrations, does not induce an inflammatory effect in macrophages.

As the next step, the potential anti-inflammatory effect of Amisulpride treatment on macrophages in an inflammatory environment stimulated via LPS-containing culture conditions was measured. Expectedly, different doses of Amisulpride affected the macrophages at varying levels, suggesting

a general anti-inflammatory effect. However, not all doses exhibited a similar impact on the inflammatory profile in terms of inhibitory response strength (Fig. 3).

LPS in the culture medium was shown to create an inflammatory environment by increasing pro-inflammatory TNF- α levels compared to the control group. However, Amisulpride treatment caused a noticeable decrease in TNF- α levels. Furthermore, increasing Amisulpride doses resulted

in a significant, dose-dependent suppression of TNF- α secretion (Fig. 3A). Similar to TNF- α , IL-6 was also progressively suppressed in proportion to the increasing dose, counteracting the inflammatory activity induced by LPS stimulation (Fig. 3B). GM-CSF levels, by contrast, exhibit slight but not significant suppression with lower concentrations of Amisulpride treatment. However, the highest dose of Amisulpride (10 $\mu\text{g/mL}$) effectively reduces GM-CSF secretion. When inter-dose variations were considered, it was noted that the 10 $\mu\text{g/mL}$ Amisulpride dose significantly suppressed GM-CSF secretion more than the 1 $\mu\text{g/mL}$ dose in an LPS-induced inflammatory environment (Fig. 3C). Under baseline conditions, Amisulpride treatment did not increase IL-12p40 secretion. Conversely, under LPS-containing inflammatory culture conditions, Amisulpride significantly reduced the elevated IL-12p40 levels. A detailed comparison indicated that IL-12p40 secretion significantly decreased with all applied Amisulpride doses, and this reduction was further enhanced with increasing concentrations (Fig. 3D).

These findings revealed that the well-known antipsychotic agent Amisulpride does not exhibit an immunostimulatory effect mediated by macrophage cells. However, under LPS-induced inflammatory conditions, Amisulpride was found to be effective in suppressing macrophage-derived pro-inflammatory cytokine secretion.

Discussion

Psychiatric disorders such as schizophrenia (SCH), bipolar disorder (BP), and major depressive disorder (MDD) have been associated with abnormalities in immune dysregulation [21, 28–32]. This association underscores the importance of investigating the immunomodulatory effects of antipsychotic agents commonly used in their treatment [17]. Although numerous reports have addressed the effects of antipsychotic drugs on immunity, research on the impact of amisulpride on the immune system is limited [33–36]. Moreover, to our knowledge, the immunomodulatory effects of amisulpride mammalian macrophage cells have not been specifically studied.

In this study, we investigated the immunomodulatory effects of Amisulpride on J774.4 macrophage cells *in vitro*. J774.4 cells were treated with 1, 5, and 10 $\mu\text{g/mL}$ Amisulpride for 24 h. Under these conditions, none of the doses applied exhibited significant effects on viability, and the detected levels of proinflammatory cytokines TNF- α , GM-CSF, IL-6, and IL-12p40 remained unchanged compared to the control conditions, suggesting that Amisulpride does not induce inflammatory responses at these non-toxic concentrations. To further assess the impact of Amisulpride under

inflammatory conditions, the cells were stimulated with lipopolysaccharide (LPS) to induce a proinflammatory state. Following LPS induction, Amisulpride treatment resulted in a dose-dependent anti-inflammatory response. Notably, the inhibition of the secretion levels of various proinflammatory cytokines exhibited differential modulation depending on the Amisulpride concentration applied.

Tumor necrosis factor-alpha (TNF- α), a pro-inflammatory cytokine primarily secreted by M1 macrophages, plays an integral role in immune-inflammatory responses [37]. M1 macrophage activation, involved in promoting inflammation in schizophrenia, primarily through TNF- α secretion during both first-episode psychosis and acute episodes, including those with comorbid depression [38]. Additionally, persistent TNF- α elevation, which is often unresponsive to antipsychotics in chronic and treatment-resistant schizophrenia, underscores its role in sustained inflammatory signaling, suggesting a pathological mechanism linked to immune dysregulation [39]. Our data demonstrated that Amisulpride treatment at non-toxic doses (1, 5, and 10 $\mu\text{g/mL}$) did not induce TNF- α secretion from macrophage cells. However, under LPS-induced inflammatory conditions, Amisulpride significantly reduced the elevated TNF- α secretion by macrophages. Moreover, TNF- α levels in the milieu decreased in a dose-dependent manner with increasing Amisulpride concentrations, supporting the anti-inflammatory effect of antipsychotics on TNF- α levels. The reduction in TNF- α secretion following Amisulpride treatment in our findings aligns with the lowered TNF- α levels reported in acute exacerbation of schizophrenia patients receiving antipsychotic therapy [40].

IL-6 is a key mediator of inflammation with pleiotropic properties. It can exhibit both pro-inflammatory and anti-inflammatory effects, which vary depending on the immune response environment [41]. Previous studies have indicated that antipsychotic drugs such as risperidone and clozapine exert anti-inflammatory effects by altering IL-6 secretion from macrophages stimulated with LPS [42]. In our results, Amisulpride exposure at three different concentrations did not induce IL-6 secretion; however, LPS-induced inflammatory conditions did. Although the 1 $\mu\text{g/mL}$ and 5 $\mu\text{g/mL}$ doses of Amisulpride significantly inhibited IL-6 secretion under inflammatory conditions, there was no significant difference between them. Moreover, the highest dose (10 $\mu\text{g/mL}$) of Amisulpride exhibited a significantly greater anti-inflammatory effect compared to the lower doses. These findings are consistent with previous studies demonstrating the anti-inflammatory effects of Amisulpride, particularly its ability to reduce IL-6 levels under inflammatory conditions [43–45]. Indeed, this reduction showed a negative correlation with schizophrenia-associated behaviors in rats, highlighting its potential therapeutic effects [43]. While

Amisulpride exposure did not induce IL-6 secretion in basal conditions, its inhibitory effect on LPS-induced IL-6 secretion supports the suggestions that Amisulpride may exert anti-inflammatory actions by modulating pro-inflammatory signaling pathways, in line with the literature [46].

Granulocyte–macrophage colony-stimulating factor (GM-CSF), a member of the colony-stimulating factor family, plays a crucial role in the survival, differentiation, and maturation of monocytic cells by acting as both a growth factor and a proinflammatory cytokine [47]. Additionally, it exhibits immunomodulatory functions, capable of promoting or suppressing immune responses in a concentration-dependent manner [48, 49]. Our results identified a distinct pattern of GM-CSF secretion under LPS-stimulated inflammatory conditions. Given that GM-CSF has been linked to several autoimmune diseases and psychosis risk, its increased expression upon LPS-stimulated inflammatory conditions supports its involvement in pro-inflammatory conditions [48, 50]. Contrary to our findings, GM-CSF levels were found to be increased with clozapine treatment, another atypical antipsychotic, under both sole treatment and LPS exposure [51, 52]. Although both Amisulpride and Clozapine are antipsychotics, their divergent effects on GM-CSF secretion and inflammation in these studies likely reflect their distinct pharmacological profiles. While Clozapine is associated with enhanced inflammation and metabolic dysregulation specifically in insulin-responsive cells, Amisulpride appears ineffective on mitochondrial functions, suggesting that these effects are cell-type and drug-specific rather than universal to all antipsychotics [51, 53].

Notably, only the highest concentration of Amisulpride (10 µg/mL) significantly reduced GM-CSF secretion, whereas lower concentrations (1 and 5 µg/mL) were insufficient to produce a similar inhibitory effect. This indicates that the anti-inflammatory conditions sufficient for reducing IL-6, TNF- α , and IL-12p40 levels were inadequate to suppress GM-CSF production (Fig. 3). The inhibitory effect of amisulpride on GM-CSF levels in our results is in line with the reduced GM-CSF levels observed with risperidone, another atypical antipsychotic treatment [54]. However, to the best of our knowledge, there are no studies directly investigating GM-CSF changes upon amisulpride use under inflammatory conditions.

IL-12p40 is a shared subunit of the proinflammatory cytokines IL-12 and IL-23 both play a critical role in the autoimmune inflammatory response. Elevated IL-12/23p40 levels in serum have been proposed as an immune phenotype indicator for psychosis patients with metabolic syndrome and have been suggested as a predictive marker for psychosis risk among high-risk groups [55, 56]. Consistent with these, our findings demonstrated that LPS-stimulated

inflammation increased IL-12p40 secretion from macrophage cells. In our results, Amisulpride exposure did not trigger inflammation; rather, it exhibited an anti-inflammatory effect, as evidenced by the decreased secretion of IL-12p40 in LPS-stimulated medium. Moreover, all three tested concentrations effectively reduced IL-12p40 levels, with statistically significant reductions observed among the concentrations. Studies proposing the elevation of IL-12 as a potential trait indicator for schizophrenia and its role in disease pathogenesis have been documented [24, 57]. Furthermore, Guo and colleagues, who investigated immune phenotypes in schizophrenia patients receiving antipsychotic treatment for at least one year, reported that IL-12p40 levels were not significantly different from those observed in healthy individuals [25]. Although divergent findings have been reported in the literature, the inhibitory effect of Amisulpride on IL-12p40 observed in our study suggests that it may provide insights into the ongoing inflammatory processes during schizophrenia treatment.

Collectively, the findings from this study indicate that although Amisulpride does not induce inflammatory responses under basal conditions, it exhibits a notable anti-inflammatory effect on the secretion of pro-inflammatory cytokines IL-12p40, IL-6, TNF- α , and GM-CSF, suggesting its immunomodulatory properties.

However, certain limitations of this study should be acknowledged to provide a more comprehensive interpretation of the findings. The cytokine panel employed in this study comprised a limited set of pro-inflammatory mediators (TNF- α , IL-6, GM-CSF, and IL-12p40). The lack of assessment of additional anti- and pro-inflammatory cytokines may have constrained the evaluation of amisulpride's effects on other pathways and their interactions. IL-12p40, being a common subunit of both IL-12 and IL-23, does not provide conclusive results on its own regarding the pro-inflammatory properties of these cytokines. Although many studies in the literature have focused on the p40 structure, it is evident that evaluating IL-12 and IL-23 separately may be crucial, especially in terms of targeting specific pro-inflammatory pathways. The use of only macrophage cells in this study implies that the findings are specific to this cell type. Additionally, our model in the manuscript reveals the acute (24-h) effects of amisulpride on macrophage cells. While *in vitro* experiments are invaluable for providing refined data, it should be noted that these models may not fully capture the complexity of interactions, given the systemic effects of psychosis and the ability of antipsychotics to cross the blood–brain barrier and interact with peripheral tissues [58, 59]. Further studies employing longer exposure periods and *in vivo* models will be necessary to evaluate the extent and duration of this immunomodulatory effect comprehensively.

Conclusion

Amisulpride is a second-generation antipsychotic widely used to treat psychosis, known for its selective antagonistic effects on dopamine D2/D3 receptors. In this study, the effects of Amisulpride on inflammatory mechanisms were evaluated, demonstrating that the drug does not induce inflammation under basal conditions; instead, it exhibits a significant anti-inflammatory effect under inflammatory conditions. Specifically, the ability of Amisulpride, applied at various concentrations, to significantly reduce the levels of IL-12p40, TNF- α , IL-6, and GM-CSF suggests its potential immunomodulatory properties.

These findings indicate that Amisulpride should be investigated not only for its antipsychotic effects but also for its immunomodulatory potential. The suppression of inflammatory responses may be one of the underlying mechanisms contributing to the clinical efficacy of Amisulpride in psychotic disorders, particularly those associated with inflammatory processes. This study provides a valuable contribution to the limited literature addressing the relationship between antipsychotic and immunomodulatory effects.

A more comprehensive investigation of Amisulpride's effects on inflammatory processes could enhance our understanding of its therapeutic efficacy in neuropsychiatric disorders and may lead to the development of novel treatment strategies. Future in vivo and clinical studies are required to validate these findings and further explore their clinical relevance.

Authors' contributions M.B.G., N.E.G., E.A., and F.A. conceptualized the study. M.B.G., N.E.G., E.A., F.S.Ö., B.G., B.N.Ö. and F.A. conducted the experiments. M.B.G., N.E.G., E.A and F.A. analyzed the data, created the figures and wrote the manuscript with equal contributions. All authors reviewed the manuscript before submission and approved the final version of it.

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Data availability Data will be available by the authors upon reasonable request.

Declarations During the preparation of this work, the authors used ChatGPT-4o to improve the language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

Ethical approval Not applicable.

Competing interests The authors declare that they do not have financial or non-financial conflict of interest.

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